Regional Inflation Expectations and the New Keynesian Phillips Curve

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Abstract

The U.S. Phillips curve estimated from cross-sectional variation across states has been flat for four decades. This conclusion that rests on the assumption that inflation expectations do not vary systematically across regions – an assumption that evidence from behavioral macroeconomics contradicts. We provide the first systematic attempt to relax it. We construct novel synthetic measures of household inflation expectations for U.S. metropolitan areas, combining Michigan Survey of Consumers microdata with regional demographic distributions from the American Community Survey. To address the endogeneity of expectations, we build a shift-share instrument from predetermined demographic shares and national cell-level expectation shocks. Estimating the Phillips curve across 21 metropolitan areas, we find that the roles of the expectation and unemployment channels reverse once their joint endogeneity is addressed. Before COVID (2004–2020), we estimate a flat Phillips curve and a strong expectation channel. After COVID (2021–2022), the pattern inverts: the expectations channel becomes small and insignificant, and a steep unemployment-inflation tradeoff re-emerges.

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1 Introduction

The U.S. Phillips curve, estimated from cross-sectional variation across states, has been flat for forty years (Hazell et al., 2022). That conclusion depends on a strong identifying assumption: inflation expectations do not vary systematically across regions, so time fixed effects absorb them. A parallel literature in household finance and behavioral macroeconomics documents precisely the opposite. Malmendier and Nagel (2016) show that lifetime inflation experiences shape individual expectations through experience-based learning, generating persistent heterogeneity that tracks regional economic histories. Hajdini et al. (2022) document stable differences in inflation expectations across U.S. metropolitan areas, and Goldfayn-Frank and Wohlfart (2020) and D'Acunto et al. (2021) provide further evidence that local price environments and demographic composition produce systematic regional divergence. The expectation component of inflation is therefore not common across regions, and the post-COVID surge of 2021–2023 has put the small-slope result under fresh empirical pressure.

If regional deviations in inflation expectations are correlated with regional labor-market conditions – a plausible outcome under experience-based learning, since metropolitan areas with persistently tight labor markets accumulate higher inflation expectations through repeated exposure to low unemployment and rising wages – then omitting regional expectations generates omitted-variable bias in the slope estimate. Whether that bias is large enough to change the central finding of Hazell et al. (2022) is an empirical question, and the answer matters for the design of monetary policy.

This paper provides the first systematic attempt to relax this assumption. We construct novel synthetic measures of household inflation expectations at the metropolitan-area level by combining microdata from the Michigan Survey of Consumers (MSC) with regional demographic distributions from the American Community Survey (ACS). Our two-step approach estimates the relationship between individual inflation expectations and observable characteristics at the national level, where the MSC is representative, and projects these estimates onto MSA-level distributions to generate fitted expectation series for each metropolitan area. This procedure sidesteps the small-sample limitations that have prevented the construction of reliable local expectation series from direct survey aggregation. With approximately 500 respondents per month nationwide, MSA-level subsamples of the MSC are far too small to support stable direct estimates. We develop three measures that differ in the information set used in the first stage: demographic composition alone (education, gender, marital status, and age); local house-price expectations alone; a combination of the two.

Our second methodological contribution is the construction of a shift-share instrumental variable for

regional inflation expectations that exploits cross-MSA variation in demographic composition. Inflation expectations are endogenous because they are formed using contemporaneous economic information and may therefore co-move with the same shocks that affect realized inflation (Mavroeidis et al., 2014). Expectations may also respond directly to inflation itself. To isolate variation in expectations that is plausibly exogenous to local inflationary conditions, we construct the instrument using predetermined education-by-gender exposure shares. Specifically, for each MSA m, we compute $\omega_{c,m,0}$, defined as the beginning-of-sample population share in demographic cell c. The shift component consists of national cell-level movements in inflation expectations, $\widehat{\pi}^e_{c,t}$, estimated using a Bayesian hierarchical model on microdata from the MSC. The resulting instrument combines fixed local demographic exposures with aggregate shifts in expectations across demographic groups. At any point in time, cross-MSA variation in the instrument is driven solely by differences in predetermined demographic shares, while time variation arises exclusively from national demographic-level expectation shocks.

We estimate the Phillips curve leveraging monthly variation in inflation and unemployment rates across 21 MSAs, extending the framework developed in Cerrato and Gitti (2022)[1]. Three findings emerge from the estimation. In the pre-COVID window (2004m1–2020m2), OLS recovers a standard downward-sloping Phillips curve with positive and statistically significant expectations. Instrumenting both regressors with our shift-share strategy raises the expectations coefficient between 1.6 and 3.3 times the OLS, while driving the unemployment coefficient to a small and statistically insignificant value in every specification. The pre-COVID Phillips curve is therefore flat once the joint endogeneity of expectations and unemployment is accounted for, in line with the literature. In the post-COVID window (2021m3–2022m6) the structure inverts: under the IV strategy, the expectations coefficient becomes small and statistically insignificant, while the unemployment coefficient becomes large and strongly negative, ranging from -1.358 to -1.666 across the three measures. The regional Phillips curve appears to have re-emerged during the post-pandemic inflation surge, while the apparent role of inflation expectations is weak once endogeneity is addressed, suggesting that any OLS correlation is likely driven in part by reverse causality from realized inflation to expected inflation.

This paper contributes to three strands of the literature. First, the nascent literature on regional inflation expectations. While a large body of work studies expectations at the national level (Carroll, 2003; Coibion and Gorodnichenko, 2015; Candia et al., 2024), the regional dimension has received far less

[1] In the empirical specification of Cerrato and Gitti (2022), headline inflation is regressed on MSA and time fixed effects, the unemployment rate, and the relative price of intermediate inputs. The unemployment rate is instrumented with a Bartik-like shock, and the relative price of intermediate inputs is instrumented with its 24-month lag. The specification comes from the two-region New Keynesian model with a vertically-linked production structure.

attention, in large part because existing surveys are not designed to be representative at sub-national geographies. Our synthetic-estimation approach provides a scalable solution to this data limitation and produces measures that are both geographically granular and temporally rich. Second, the literature that leverages regional panel variation to estimate the Phillips curve (Mavroeidis et al., 2014; Babb and Detmeister, 2017; Beraja et al., 2019; McLeay and Tenreyro, 2020; Hooper et al., 2020; Hazell et al., 2022; Cerrato and Gitti, 2022; Fitzgerald et al., 2024; Gitti, 2024). We are the first to estimate the regional Phillips curve while directly controlling for and instrumenting for regional inflation expectations, rather than subsuming their variation into fixed effects or the slope coefficient. Third, this work speaks to the literature on the role of expectations in the post-COVID inflation surge (Coibion and Gorodnichenko, 2025; Benigno and Eggertsson, 2023; Beaudry et al., 2024), and to a complementary body of evidence on geographically heterogeneous expectation formation (Malmendier and Nagel, 2016; Goldfayn-Frank and Wohlfart, 2020; D'Acunto et al., 2021; Binder et al., 2024; Hajdini et al., 2022).

The remainder of this paper is organized as follows. Section 2 describes the data. Section 3 develops the three synthetic expectation measures, constructs the shift-share instrument, and specifies the regional Phillips curve. Section 4 presents the empirical results across all three measures and across the pre- and post-COVID windows. Section 5 concludes.

2 Data

We assemble a monthly panel of $M = 21$ U.S. metropolitan statistical areas (MSAs) covering January 2005 through January 2025. The panel combines four data sources: household inflation expectations from the Michigan Survey of Consumers, regional demographic distributions from the American Community Survey, regional consumer-price and unemployment data from the Bureau of Labor Statistics, and house-price data from the Federal Housing Finance Agency. Each is described in turn below.

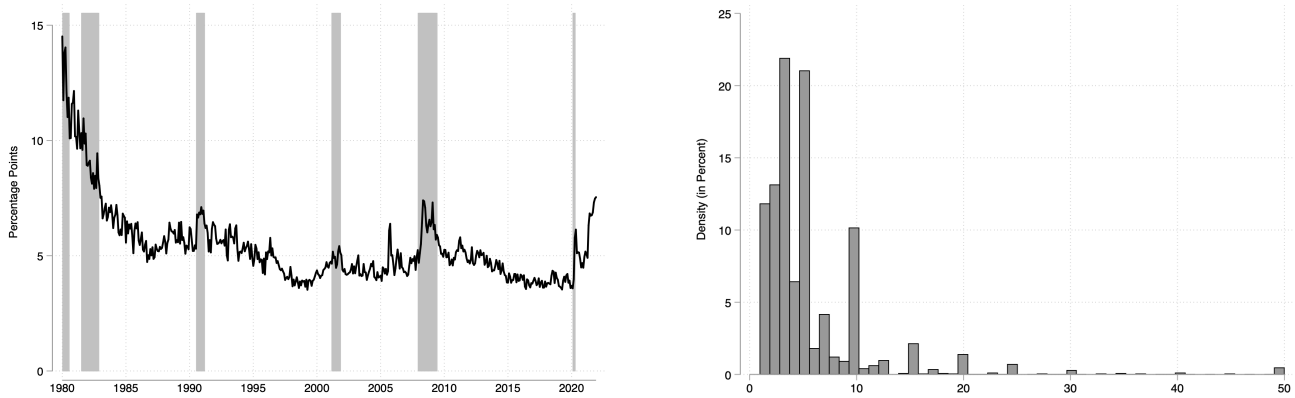
2.1 Household Inflation Expectations

We measure household inflation expectations using microdata from the Michigan Survey of Consumers (MSC). The MSC began in 1978 and is the longest-running monthly survey of household expectations in the United States. Each month it interviews approximately 500 households about expected price changes over both one-year and five-year horizons, along with demographic characteristics and geographic identifiers. We focus on the one-year-ahead point inflation expectation elicited by the question *"By about*

what percent do you expect prices to go up or down on the average, during the next 12 months?" The variable, denoted $\pi^e_{i,t}$ for respondent i in month t, takes integer values in $[0, 100]$ and is right-skewed with a small fraction of large outliers.

The MSC is designed to be representative at the *national* level. The roughly 500 respondents per month support precise national aggregates but are too thinly spread across MSAs to support direct regional aggregation: many metropolitan areas have only a handful of MSC respondents in a given month, and some have none. Simple within-MSA averages of MSC responses are therefore both noisy and unrepresentative of the underlying regional population. Section 3 develops the two-step synthetic-estimation strategy we use to bypass this limitation.

Figure 1: One-year ahead inflation expectations



(a) Time Series (b) Distribution

Notes: Panel (a) plots the sample-weighted national monthly mean of one-year-ahead inflation expectations $\pi^e_{i,t}$ in the MSC. Panel (b) plots the pooled distribution of individual responses over the sample period. MSC survey weights used throughout.

Figure 1 summarizes the time-series and cross-sectional behavior of $\pi^e_{i,t}$. The national mean was elevated in the early 1980s, fell rapidly through the Volcker disinflation, and remained stable in the 1%–5% range for most of the post-Greenspan era, with marked spikes during recessions and the post-COVID inflation surge. The bulk of the cross-sectional distribution lies between zero and ten percent, with a thin upper tail.

Summary statistics for the demographic characteristics used in our first-stage construction of regional expectations appear in Table 1. The sample is balanced across gender (54% female), skewed toward higher educational attainment (28% with at least some college), and approximately uniform across income terciles.

Table 1: Summary statistics: Michigan Survey of Consumers

Variable	MSC
Age (years)	49
Female	54%
Low income	43%
Middle income	33%
High income	25%
< High school	4%
High school	6%
Some college	28%
N (respondent-months)	277,160

Notes. Summary statistics for the demographic variables in the MSC microdata used in the first stage of our synthetic-expectation construction. Sample weighted by MSC survey weights.

2.2 Regional Demographics

Regional demographic distributions are constructed from the American Community Survey (ACS) microdata accessed through IPUMS-USA. The ACS provides annual person-level samples designed to be representative at the MSA level, with roughly one percent of the U.S. population surveyed each year. We use the 2005–2024 vintages and harmonize variables across the 2005–2021 IPUMS extract and the 2022–2024 extract.

For each MSA-year we compute population-weighted shares of the same demographic characteristics used in the MSC first stage: educational attainment in three buckets (less than high school through high-school graduate, some college through bachelor's degree, and any graduate education), gender, marital status in four categories (married, divorced, widowed, never married), and continuous age. The cell-by-MSA population shares $\omega_{c,m,0}$ that enter our shift-share instrument (Section 3.2) are constructed from the same ACS microdata.

The ACS releases data at the annual frequency; we hold demographic shares constant within calendar year when joining to monthly inflation and unemployment series.

2.3 Regional Inflation, Unemployment, and House Prices

Regional CPI inflation is constructed from the Bureau of Labor Statistics metropolitan-area CPI series. The BLS publishes monthly or bi-monthly all-items CPI indices for 21 MSAs starting in January 1988. We interpolate the bi-monthly series to a monthly frequency where necessary and compute year-over-year inflation $\pi_{m,t}$ as the twelve-month log difference of the MSA-level index. This series is the dependent variable in all Phillips-curve regressions in Section 4.

Regional unemployment $u_{m,t}$ is obtained from the BLS Local Area Unemployment Statistics (LAUS) program, which produces monthly seasonally-adjusted unemployment rates for each MSA from January 1990 onward. We use the labor-force-weighted unemployment rate for the MSA boundary that matches the corresponding CPI release.

MSA-level house-price growth is computed from the Federal Housing Finance Agency all-transactions house-price index (FHFA HPI). The index is released quarterly for each of our 21 MSAs; we interpolate linearly to a monthly frequency and define the housing-inflation series $\pi^h_{m,t}$ as the year-over-year log change. Reliable values are available from January 2007 onward.

2.4 Sample and Geographic Coverage

Our baseline sample is a balanced monthly panel of 21 MSAs from January 2005 through December 2023. Specifications that include the housing channel are restricted to January 2007 through December 2023 to match the availability of the FHFA index. The 21 MSAs are those with continuous BLS CPI coverage over this window.[2]

3 Methodology

Our empirical strategy proceeds in three steps that confront the three challenges of estimating the regional Phillips curve. The first challenge is measurement: no existing data source provides reliable, high-frequency measures of inflation expectations at the metropolitan area level. Section 3.1 develops three synthetic measures of household inflation expectations at the MSA level, all built on a common two-step procedure that projects nationally estimated demographic gradients onto MSA-level demographic

[2]The 21 MSAs are New York, Los Angeles, Chicago, Houston, Philadelphia, Phoenix, San Francisco, Boston, Dallas, Atlanta, Miami, Washington DC, Seattle, Minneapolis, San Diego, Denver, Tampa, St. Louis, Baltimore, Detroit, and Honolulu.

distributions. The second challenge is identification: a credible estimate of the expectations channel requires an instrument for $E_{m,t}[\pi_{t+1}]$ that exploits cross-sectional variation orthogonal to local supply and demand shocks. Section 3.2 constructs such a shift-share instrument from the interaction of national cell-level expectation dynamics and regional demographic composition. The third challenge is the joint identification of expectations and unemployment in the Phillips-curve regression. Section 3.3 embeds our expectations measures and instrument in a regional New Keynesian Phillips curve and discusses the auxiliary unemployment instrument and intermediate-input price control that, together with the expectations IV, deliver the slope estimate.

3.1 Constructing Regional Inflation Expectations

A central challenge for any empirical study of regional inflation dynamics is the absence of representative, high-frequency measures of inflation expectations at the metropolitan-area level. The Michigan Survey of Consumers is the longest-running and most widely used household expectation survey in the United States, but it is designed to be nationally representative. With approximately 500 respondents per month, regional subsamples are too small to support reliable direct aggregation: many MSAs have only a handful of observations in a given month, and some have none. Simple within-MSA averages of MSC responses are therefore both noisy and unrepresentative of the underlying regional population.

We address this measurement problem with a two-step synthetic-estimation procedure. In the first step, we use the full national MSC microdata to estimate the relationship between individual inflation expectations and observable demographic, socioeconomic, and local economic characteristics. In the second step, we project these national estimates onto MSA-level distributions of those same characteristics, obtained from the ACS and from administrative housing-market data, to produce fitted expectation series for each metropolitan area in each period. This approach trades the noise of direct regional aggregation for the stability of national estimates while preserving cross-MSA heterogeneity through differences in regional composition.

We develop three variants of the synthetic measure that differ in the information set used in the first stage. *Measure 1*, described in Section 3.1.1, uses demographic characteristics alone—education, gender, marital status, and age—and captures the persistent cross-sectional heterogeneity in expectations that arises from the demographic composition of each MSA. *Measure 2* (Section 3.1.2) replaces demographics with realized local house-price inflation, isolating the housing channel to inflation expectations that the household-finance literature has emphasized (Hajdini et al., 2022; D'Acunto et al., 2021). *Measure 3*

(Section 3.1.3) combines both channels: the demographic gradient and the housing channel enter the first stage jointly. We use all three measures in the Phillips-curve estimation of Section 4 to assess the robustness of our findings to the choice of information set.

3.1.1 Measure 1: Demographic-Only Regional Expectations

In the first stage we estimate the following cross-sectional model of individual inflation expectations on observable demographic characteristics:

$$E_{i,t}[\pi_{t+1}] = \beta_{0t} + \beta_{1t}\,\text{Education}_i + \beta_{2t}\,\text{Gender}_i + \beta_{3t}\,\text{MaritalStatus}_i + \beta_{4t}\,\text{Age}_i + \varepsilon_{i,t}. \tag{1}$$

Here $E_{i,t}[\pi_{t+1}]$ denotes the one-year-ahead inflation expectation of MSC respondent i surveyed in month t. Education, Gender, and MaritalStatus are categorical, and Age is continuous. All β coefficients are indexed by t: the model is estimated *separately* in each month so that the coefficients capture time-varying shifts in the demographic gradient of expectations at the national level. The estimation uses MSC survey weights.

In the second stage we project the estimated coefficients onto MSA-level demographic distributions obtained from the ACS:

$$\widehat{E}_{m,t}[\pi_{t+1}] = \widehat{\beta}_{0t} + \widehat{\beta}_{1t}\,\overline{\text{Education}}_{m,t} + \widehat{\beta}_{2t}\,\overline{\text{Gender}}_{m,t} + \widehat{\beta}_{3t}\,\overline{\text{MaritalStatus}}_{m,t} + \widehat{\beta}_{4t}\,\overline{\text{Age}}_{m,t}, \tag{2}$$

where the overline denotes MSA-level demographic means computed from the ACS for each year. ACS shares are annual; first-stage coefficients are monthly. The resulting synthetic measure $\widehat{E}_{m,t}[\pi_{t+1}]$ combines two distinct sources of variation: cross-sectional variation arising from persistent differences in demographic composition across MSAs, and time-series variation arising from monthly shifts in the national demographic gradient $\widehat{\beta}_t$. By construction, the measure bypasses the small-sample and representativeness problems of direct regional aggregation: the identifying information is national, projected onto census-quality regional demographic data.

3.1.2 Measure 2: Housing-Only Regional Expectations

Inflation expectations are shaped not only by who a household is, but by where it lives. Households in regions experiencing rapid house-price appreciation tend to report higher inflation expectations, reflecting the salience of housing costs in their consumption basket and direct experience of local price pressures

(Hajdini et al., 2022; D'Acunto et al., 2021). Measure 1 is silent on this channel. Measure 2 replaces the demographic gradient with realized local house-price inflation, isolating the housing channel.

In the first stage we estimate

$$E_{i,t}[\pi_{t+1}] = \beta_{0t} + \beta_{1t}\, E_{i,t}[\pi^h_{t+1}] + \varepsilon_{i,t}, \tag{3}$$

where $E_{i,t}[\pi^h_{t+1}]$ denotes individual i's one-year-ahead house-price expectation, elicited directly in the MSC. The coefficient β_{1t} captures the time-varying pass-through from housing expectations to overall inflation expectations at the national level. No demographic controls enter the first stage of Measure 2.

Individual house-price expectations are not available at the MSA level in the ACS. In the second stage we therefore replace $E_{i,t}[\pi^h_{t+1}]$ with realized MSA-level house-price inflation, $\pi^h_{m,t}$, defined as the year-over-year log change of the FHFA all-transactions house-price index:

$$\widehat{E}_{m,t}[\pi_{t+1}] = \widehat{\beta}_{0t} + \widehat{\beta}_{1t}\, \pi^h_{m,t}. \tag{4}$$

The substitution is motivated by the high serial correlation of local housing markets: MSA-level house-price growth is highly persistent, so realized housing inflation is a strong predictor of one-year-ahead expectations (Binder et al., 2024). The FHFA index is quarterly; we interpolate linearly to a monthly frequency. The Measure 2 sample is restricted to 2007–2024 to match the availability of the FHFA index.

3.1.3 Measure 3: Demographic and Housing Regional Expectations

Measure 3 combines both channels by including demographic controls *and* the housing-expectation regressor in a single first stage. The first-stage specification is

$$E_{i,t}[\pi_{t+1}] = \beta_{0t} + \beta_{1t}\,\text{Education}_i + \beta_{2t}\,\text{Gender}_i + \beta_{3t}\,\text{MaritalStatus}_i + \beta_{4t}\,\text{Age}_i + \beta_{5t}\, E_{i,t}[\pi^h_{t+1}] + \varepsilon_{i,t}, \tag{5}$$

and the second-stage projection is

$$\widehat{E}_{m,t}[\pi_{t+1}] = \widehat{\beta}_{0t} + \widehat{\beta}_{1t}\,\overline{\text{Education}}_{m,t} + \widehat{\beta}_{2t}\,\overline{\text{Gender}}_{m,t} + \widehat{\beta}_{3t}\,\overline{\text{MaritalStatus}}_{m,t} + \widehat{\beta}_{4t}\,\overline{\text{Age}}_{m,t} + \widehat{\beta}_{5t}\,\pi^h_{m,t}. \tag{6}$$

As with Measure 2, the realized MSA-level house-price inflation $\pi^h_{m,t}$ substitutes for the unobserved individual house-price expectation in the second stage. The Measure 3 sample is likewise restricted to 2007–2024 to match the FHFA availability.

The three measures span a natural trade-off between informational content and the population-level interpretability of the first-stage gradient. We use all three in the Phillips-curve estimation of Section 4.

3.2 A Shift-Share Instrument for Regional Inflation Expectations

The synthetic measures constructed in Section 3.1 solve the measurement problem but are not yet causally identified. The baseline measure projects national gradients onto MSA-level demographics, and the local-area measure additionally incorporates realized regional house-price inflation, both of which co-move with realized regional CPI inflation through shared demand and supply shocks. They may also suffer from reverse causality, responding to inflation itself. To identify the expectations channel in the regional Phillips curve we therefore construct a shift-share instrumental variable for $E_{m,t}[\pi_{t+1}]$ in the spirit of Bartik (1991) and Borusyak et al. (2022).

3.2.1 Demographic Cells

We partition the adult U.S. population into $C = 6$ demographic cells defined by the interaction of educational attainment and gender. Education is collapsed into three buckets—*less-than-college* (less than high school through high-school graduate), *some-college* (some college through bachelor's degree), and *graduate* (any post-baccalaureate education). Gender takes two values, male and female. The same six-category partition is applied identically in both data sources: MSC microdata, where cells index respondents, and ACS microdata, where cells index residents in each MSA. Because the MSC and IPUMS-ACS use different raw education codings, we harmonize the two at the bucket level so that the label "some-college" carves the same population in both datasets.

The partition is motivated by the household-finance literature on heterogeneous expectation formation. D'Acunto et al. (2021) document that exposure to grocery prices shifts inflation expectations through information environments that differ by education. Hajdini et al. (2022) show that household inflation expectations vary systematically with education and demographic characteristics. The three-bucket coarsening is the deepest education partition the MSC monthly sample supports without producing pervasively empty cell-month bins; cross-tabulated with gender it yields six cells with sufficient density in both surveys. A parallel age-by-gender partition is also constructed for use in robustness checks not reported here.

3.2.2 National Cell-Level Expectations

We recover cell-time inflation expectations $\widehat{\pi}^e_{c,t}$ by fitting a national-only Bayesian hierarchical model on the MSC microdata,

$$\pi^e_{i,t} = \alpha_t + \mu_{c[i],t} + X_i'\zeta + \nu_{i,t}, \tag{7}$$

$$\mu_{c,t} \mid \sigma^2_\mu \sim \mathcal{N}\big(0, \sigma^2_\mu\big)\,, \tag{8}$$

$$\alpha_t \mid \sigma^2_\alpha \sim \mathcal{N}\big(0, \sigma^2_\alpha\big)\,, \tag{9}$$

$$\nu_{i,t} \sim \mathcal{N}\big(0, \sigma^2_\nu\big)\,, \tag{10}$$

where $\pi^e_{i,t}$ is respondent i's one-year-ahead point inflation expectation in month t, α_t is a time random effect, $\mu_{c,t}$ is a cell-time hierarchical random effect, and X_i collects additive individual controls (marital status, homeownership, and continuous age). Variance components σ^2_μ, σ^2_α, and σ^2_ν receive weakly informative half-normal priors on the standard-deviation scale, following Gelman (2006). The cell-time prediction is the posterior mean

$$\widehat{\pi}^e_{c,t} = \mathbb{E}\big[\alpha_t + \mu_{c,t} \,\big|\, \text{data}\big]\,. \tag{11}$$

By construction, $\widehat{\pi}^e_{c,t}$ pools information across cells and across time periods through the hierarchical priors, shrinking thinly-sampled cell-month bins toward the time-specific national mean in the spirit of Park et al. (2004) and Ghitza and Gelman (2013).

Crucially, equation (7) carries no MSA-level effects: the model is national-only, and the cell-time predictions $\widehat{\pi}^e_{c,t}$ contain no MSA-specific information. This separation is what allows the resulting instrument to satisfy the exclusion restriction described below.

3.2.3 Regional Cell Shares

For each MSA m, 2005 ACS microdata yield ACS-weighted population counts $N_{c,m}$ by cell fixed at the beginning of the sample. The cell share within an MSA is

$$\omega_{c,m,0} = \frac{N_{c,m,0}}{N_{m,0}}, \qquad N_{m,0} = \sum_{c=1}^{C} N_{c,m,0}. \tag{12}$$

The shares $\omega_{c,m,0}$ encode the local education-gender composition of MSA m at time 0 and sum to one within each $(m, 0)$.

3.2.4 Shift-Share Instrument

The expectations instrument is the inner product of national cell-time predictions and regional cell shares,

$$Z^{\pi}_{m,t} = \sum_{c=1}^{C} \omega_{c,m,t}\, \widehat{\pi}^{e}_{c,t}. \tag{13}$$

$Z^{\pi}_{m,t}$ inherits its time variation from the national objects $\{\widehat{\pi}^{e}_{c,t}\}$ (the "shifts") and its cross-sectional variation from the regional shares $\{\omega_{c,m,t}\}$ (the "shares"). Intuitively, $Z^{\pi}_{m,t}$ is the inflation expectation that MSA m would report at time t if every resident held the average national expectation of someone in their education-gender cell. Cross-MSA variation in $Z^{\pi}_{m,t}$ at time t therefore reflects *only* differences in demographic composition; the time dimension reflects *only* national cell-level expectation dynamics.

3.2.5 Identification

Identification of the expectations coefficient ψ in the regional Phillips curve (introduced in Section 3.3) rests on the exclusion restriction that the national cell-time predictions $\widehat{\pi}^{e}_{c,t}$ are uncorrelated with the regional Phillips-curve residual $\varepsilon_{m,t}$, conditional on MSA and time fixed effects, the unemployment instrument $Z^{u}_{m,t}$, and the lagged intermediate-input price control. Because $\widehat{\pi}^{e}_{c,t}$ is constructed from a national-only model with no MSA random effects, the residual identifying variation is the interaction of national cell-level expectation movements with cross-sectional cell shares, which is plausibly orthogonal to MSA-specific supply or demand shocks conditional on the included controls.

Following Goldsmith-Pinkham et al. (2020), we report the share-by-share decomposition of $Z^{\pi}_{m,t}$ to identify which education-gender cells carry the bulk of the identifying weight. As a falsification test we regress decennial Census cell shares from a pre-period on observable contemporaneous MSA characteristics and report the resulting placebo coefficients.

3.2.6 Estimation and Inference

Because $\widehat{\pi}^{e}_{c,t}$ in equation (11) is a posterior summary from a Bayesian first stage, standard two-stage least-squares inference understates uncertainty by treating the constructed instrument as observed data (Murphy and Topel, 1985; Pagan, 1984). We adopt the multiple-imputation correction of Rubin (1987).

For each of S draws from the first-stage posterior, $\widehat{\pi}_{c,t}^{e,(s)}$, we construct the draw-specific instrument

$$Z_{m,t}^{\pi,(s)} = \sum_{c=1}^{C} \omega_{c,m,t}\, \widehat{\pi}_{c,t}^{e,(s)}$$

and re-estimate the second-stage 2SLS regression, collecting the per-draw point estimate $\widehat{\psi}^{(s)}$ and within-draw variance-covariance matrix $\widehat{V}^{(s)}$. The reported point estimate is the across-draw average $\widehat{\psi} = S^{-1}\sum_s \widehat{\psi}^{(s)}$, and inference uses Rubin's combination,

$$\widehat{\mathrm{Var}}(\widehat{\psi}) = \frac{1}{S}\sum_{s=1}^{S} \widehat{V}^{(s)} + \frac{S+1}{S}\cdot\frac{1}{S-1}\sum_{s=1}^{S}\left(\widehat{\psi}^{(s)} - \widehat{\psi}\right)^2. \tag{14}$$

The two terms decompose total uncertainty into within-draw 2SLS sampling error and between-draw posterior uncertainty over the first-stage cell-time predictions.

3.3 The Regional Phillips Curve

Using the synthetic expectation measures of Section 3.1 and the shift-share instrument of Section 3.2, we estimate the following regional New Keynesian Phillips curve on a monthly panel of $M = 21$ MSAs, following Cerrato and Gitti (2022):

$$\pi_{m,t} = \eta_m + \xi_t + \psi\, E_{m,t}[\pi_{t+1}] - \kappa\, u_{m,t} + \delta\, p_{m,t-24}^{x} + \varepsilon_{m,t}, \tag{15}$$

where $\pi_{m,t}$ is the year-over-year headline CPI inflation rate in MSA m at month t, $E_{m,t}[\pi_{t+1}]$ is the one-year-ahead inflation expectation, $u_{m,t}$ is the unemployment rate[3], and $p_{m,t}^{x}$ is the relative intermediate-input price control included on the grounds that firms use both labor and intermediate inputs in production. The MSA fixed effects η_m absorb time-invariant local characteristics; the year-quarter fixed effects ξ_t absorb aggregate shocks common to all metropolitan areas, including fiscal and monetary policy responses.

As Hazell et al. (2022) show by iterating the Phillips curve forward, the time fixed effects also absorb the common long-run inflation expectation anchored to the prevailing monetary policy regime. By directly including a measure of *short-run* regional expectations alongside time fixed effects, we isolate the role of *regional heterogeneity* in expectations over and above the common aggregate component.

[3]Specifically, $u_{m,t} = -\log(1 - u_{m,t})$.

OLS estimation of equation (15) treats $E_{m,t}[\pi_{t+1}]$ as exogenous, which is implausible: any reasonable synthetic measure of regional expectations co-moves with realized regional inflation through shared demand and supply shocks, so the OLS estimate of ψ is biased. We instrument $E_{m,t}[\pi_{t+1}]$ with the shift-share variable $Z^{\pi}_{m,t}$ constructed in Section 3.2. Identification rests on the exclusion restriction discussed in Section 3.2.5: the national cell-time predictions $\widehat{\pi}^{e}_{c,t}$ are uncorrelated with $\varepsilon_{m,t}$ conditional on the fixed effects, the unemployment instrument, and the lagged relative intermediate input price control.

A consistent estimate of the Phillips-curve slope κ requires accounting for regional supply shocks that move both unemployment and inflation. Following Cerrato and Gitti (2022), we instrument $u_{m,t}$ with a Bartik shift-share instrument $Z^{u}_{m,t}$ that interacts pre-determined MSA-level employment shares in intermediate-input sectors with national sectoral labor-demand shocks, isolating variation in local unemployment driven by shifts in intermediate input sector demand.

The cost-push channel introduces a second source of simultaneity, since contemporaneous regional input prices respond to regional inflation shocks. Following Cerrato and Gitti (2022), we use the 24-month lag $p^{x}_{m,t-24}$ as a control to absorb the reverse-causation component while preserving the cost-push identification of δ.

4 Estimating the Regional Phillips Curve

We present estimates of the regional Phillips curve in equation (15) for each of the three expectation measures developed in Section 3.1. Three subsections present, in turn, results for Measure 1 (demographic-only), Measure 2 (housing-only), and Measure 3 (combined). Each table pairs OLS and 2SLS estimates of equation (15) on two sample windows: a *pre-COVID* window (January 2004 through February 2020) and a *post-COVID* window (March 2021 through June 2022) that isolates the post-pandemic inflation surge. Both windows include MSA and year-quarter fixed effects.[4]

A consistent pattern emerges across the three measures. First, in the pre-COVID window, instrumenting raises the expectations coefficient $\widehat{\psi}$ and drives the unemployment coefficient toward zero. The OLS Phillips curve is steep and significant; the IV Phillips curve is flat. Second, in the post-COVID window the pattern inverts: the unemployment coefficient becomes large and strongly negative under

[4]The 2SLS specifications replace year-quarter fixed effects with year-quarter $\times$ initial-industry-share time effects, consistent with the shift-share identification described in Sections 3.2–3.3. The expectations instrument is the education-gender shift-share $Z^{\pi}_{m,t}$ defined in equation (13); the unemployment instrument is the Bartik $Z^{u}_{m,t}$ following Cerrato and Gitti (2022); the relative intermediate input price $p^{x}_{m,t}$ is instrumented with its 24-month lag. Standard errors are clustered at the MSA $\times$ year-quarter level.

IV, while the expectations coefficient becomes small and statistically indistinguishable from zero. The IV-corrected Phillips curve appears to have steepened dramatically during the post-pandemic inflation surge, even as the expectations channel weakens.

4.1 Measure 1: Demographic-Only Expectations

Table 2 reports estimates with $\widehat{E}_{m,t}[\pi_{t+1}]$ constructed from the demographic-only first stage in equations (1)–(2).

Table 2: Phillips Curve Estimates: Michigan Survey Measure 1 (Demographic)

	Pre Period (2004–2020)		Post Period (2021m3–2022m6)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
$u_{m,t}$	-0.238***	0.209	-0.063	-1.358***
	(0.027)	(0.130)	(0.074)	(0.250)
$E_{m,t}[\pi_{t+1}]$	0.264***	0.318***	0.519***	0.127
	(0.042)	(0.069)	(0.045)	(0.106)
$p^x_{m,t}$	-0.081***	0.157***	-0.228***	0.120
	(0.013)	(0.034)	(0.038)	(0.085)
Observations	2,650	2,398	336	336
Time FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
F-stat $Z^{\pi}_{m,t}$	—	2233.31	—	365.47
F-stat $Z^{u}_{m,t}$	—	51.93	—	36.55

Notes: OLS (columns 1 and 3) and IV (columns 2 and 4) estimates of the New Keynesian Phillips Curve. The dependent variable is the MSA-level inflation rate (12-month percentage change in the CPI all-items), $\pi_{m,t}$. Inflation expectations $E_{m,t}[\pi_{t+1}]$ are constructed from Michigan Survey Measure 1 (demographic-only first stage). $u_{m,t}$ is the negative log employment rate. $p^x_{m,t}$ is the log ratio of manufacturing PPI to CPI all items. In the IV specification, $E_{m,t}[\pi_{t+1}]$ is instrumented by $Z^{\pi}_{m,t}$, the gender-by-education shift-share instrument; $u_{m,t}$ is instrumented by $Z^{u}_{m,t}$, the Bartik shift-share instrument; $p^x_{m,t}$ is instrumented by its 24-month lagged value. The OLS specification absorbs MSA and year-quarter fixed effects. The IV specification absorbs MSA fixed effects and year-quarter $\times$ initial industry-share time effects. Standard errors (in parentheses) are clustered at the MSA $\times$ year-quarter level. The pre period covers 2004m1–2020m2; the post period covers 2021m3–2022m6. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

In the pre-COVID window, OLS (column 1) recovers a standard downward-sloping Phillips curve: expectations enter with a positive significant coefficient of 0.264, $u_{m,t}$ enters with a negative significant coefficient of -0.238.Instrumenting (column 2) raises the expectations coefficient by roughly twenty percent to 0.318, indicating that OLS is biased *downward* relative to the IV-corrected estimate. The unemployment coefficient loses significance: once the endogeneity of unemployment is accounted for by the

Bartik instrument, the pre-COVID Phillips curve becomes flat, in line with the literature. The first-stage F-statistic on $Z^{\pi}_{m,t}$ and on $Z^{u}_{m,t}$ of exceed conventional weak-instrument thresholds.

In the post-COVID window the relationship changes markedly. The OLS expectations coefficient roughly doubles to 0.519 while the unemployment coefficient becomes small and insignificant (-0.063), reflecting the strong contemporaneous co-movement of inflation and unemployment expectations during the 2021–2022 surge. Under IV (column 4), expectations are no longer significant, while the unemployment coefficient becomes large and strongly negative ($\widehat{\kappa} = -1.358^{***}$). The instrumented Phillips curve has steepened by roughly an order of magnitude relative to the pre-COVID OLS estimate. First-stage F-statistics remain large.

4.2 Measure 2: Housing-Only Expectations

Table 3 reports estimates with $\widehat{E}_{m,t}[\pi_{t+1}]$ constructed from the housing-only first stage in equations (3)–(4). The sample is restricted to 2007m1 onward to match the availability of FHFA house-price data.

Table 3: Phillips Curve Estimates: Michigan Survey Measure 2 (Housing-Only)

	Pre Period (2004–2020)		Post Period (2021m3–2022m6)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
$u_{m,t}$	-0.226***	0.147	-0.021	-1.666***
	(0.028)	(0.111)	(0.070)	(0.496)
$E_{m,t}[\pi_{t+1}]$	0.088***	0.291***	0.227***	0.297
	(0.027)	(0.049)	(0.026)	(0.306)
$p^{x}_{m,t}$	-0.097***	0.114***	-0.199***	-0.118
	(0.015)	(0.032)	(0.041)	(0.316)
Observations	2,321	2,237	336	336
Time FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
F-stat $Z^{\pi}_{m,t}$	—	267.73	—	23.74
F-stat $Z^{u}_{m,t}$	—	51.93	—	36.55

Notes: Same conventions as Table 2, except that $E_{m,t}[\pi_{t+1}]$ is constructed from Michigan Survey Measure 2 (housing-only first stage; see equations (3)–(4)). The sample begins in 2007m1 to match the availability of FHFA house-price data. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The pattern parallels that of Measure 1 closely. In the pre-COVID window, the OLS expectations coefficient $\widehat{\psi}^{\text{OLS}} = 0.088$ is smaller in absolute magnitude than for Measure 1, reflecting the narrower information content of a single-channel first stage. The IV estimate $\widehat{\psi}^{\text{IV}} = 0.291$ is more than three

times the OLS, again documenting a substantial downward OLS bias. The unemployment coefficient moves from -0.226 under OLS to a positive insignificant 0.147 under IV: the pre-COVID Phillips curve is once again flat once endogeneity is addressed. In the post-COVID window the expectations coefficient is insignificant under both OLS and IV at standard levels, while the IV unemployment coefficient spikes to -1.666—the most negative across the three measures.

4.3 Measure 3: Demographic and Housing Expectations

Table 4 reports estimates with $\widehat{E}_{m,t}[\pi_{t+1}]$ constructed from the combined first stage in equations (5)–(6), which incorporates both the demographic gradient and local house-price inflation.

Table 4: Phillips Curve Estimates: Michigan Survey Measure 3 (Combined)

	Pre Period (2004–2020)		Post Period (2021m3–2022m6)	
	(1) OLS	(2) IV	(3) OLS	(4) IV
$u_{m,t}$	-0.229***	0.109	0.001	-1.469***
	(0.028)	(0.108)	(0.081)	(0.299)
$E_{m,t}[\pi_{t+1}]$	0.180***	0.382***	0.120***	0.093
	(0.040)	(0.067)	(0.019)	(0.087)
$p^x_{m,t}$	-0.101***	0.102***	-0.083**	0.140*
	(0.015)	(0.031)	(0.035)	(0.078)
Observations	2,321	2,237	336	336
Time FE	Yes	Yes	Yes	Yes
MSA FE	Yes	Yes	Yes	Yes
F-stat $Z^{\pi}_{m,t}$	—	329.03	—	88.64
F-stat $Z^{u}_{m,t}$	—	51.93	—	36.55

Notes: Same conventions as Table 2, except that $E_{m,t}[\pi_{t+1}]$ is constructed from Michigan Survey Measure 3 (combined first stage with demographic controls and local house-price inflation; see equations (5)–(6)). The sample begins in 2007m1 to match the availability of FHFA house-price data. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The combined first stage delivers the largest IV estimate of the expectations coefficient in the pre-COVID window: $\widehat{\psi}^{\text{IV}} = 0.382$, more than twice the OLS estimate of 0.180. The unemployment coefficient again moves from -0.229 under OLS to an insignificant 0.109 under IV. In the post-COVID window the IV expectations coefficient is small (0.093, insignificant) while the unemployment coefficient is large and strongly negative (-1.469).

Three observations cut across the three measures. First, the IV-corrected expectations coefficient in the pre-COVID window is remarkably stable at $\widehat{\psi}^{\text{IV}} \in [0.29, 0.38]$, substantially above all three OLS

estimates. The expectations channel is robust and economically meaningful. Second, under IV the pre-COVID unemployment coefficient is statistically indistinguishable from zero across all three measures. Correcting for endogeneity eliminates the apparent OLS Phillips curve. Third, in the post-COVID window the unemployment coefficient becomes large and strongly negative across all three measures (between -1.36 and -1.67). The IV expectations coefficient becomes small and is no longer significant. The Phillips curve appears to have re-emerged in the post-pandemic period, with the regional unemployment-inflation tradeoff sharpening even as the expectations channel attenuates.

Items flagged for further review. **[TBD: The post-COVID window has only $N = 336$ observations (approximately 21 MSAs $\times$ 16 months); we should discuss whether to extend it through later 2022 or beyond, given that some MSAs may have CPI releases beyond 2022m6.] [TBD: The sign of the unemployment coefficient flips between OLS and IV in the pre-COVID window. This is striking and deserves a sensitivity analysis: how sensitive is the flip to the choice of $Z^u_{m,t}$ (Cerrato–Gitti tradeable-demand Bartik), to the input price control, or to the year-quarter $\times$ industry-share time effects?] [TBD: The first-stage F-statistic on $Z^\pi_{m,t}$ for Measure 2 in the post-COVID window (F = 23.74) is the lowest across all specifications. Worth a brief comment on instrument strength.] [TBD-Giulia: Confirm that the post-COVID IV unemployment coefficient of -1.36 to -1.67 is consistent with the post-COVID Phillips curve literature; specifically, compare to Benigno and Eggertsson (2023) and Beaudry et al. (2024) on the post-COVID resurrection of the Phillips curve.]**

5 Conclusion

The dominant approach to estimating the regional Phillips curve (Hazell et al., 2022) identifies the slope from cross-sectional variation in unemployment and inflation, absorbing aggregate shocks and common long-run inflation expectations through two-way fixed effects. The identification rests on the assumption that inflation expectations do not vary systematically across regions. A parallel literature in household finance establishes the opposite: lifetime experience, local price environments, and demographic composition produce persistent regional heterogeneity in inflation expectations. If that heterogeneity is correlated with regional labor-market conditions, omitting it biases the slope estimate.

This paper provides the first systematic test of that hypothesis. We construct three synthetic measures of household inflation expectations at the MSA level, develop a shift-share instrument from the interaction

of national education-gender cell-level expectation dynamics with regional education-gender shares, and estimate the regional Phillips curve on a monthly panel of 21 MSAs across both a pre-COVID and a post-COVID window, instrumenting both expectations and unemployment.

Three findings stand out. First, in the pre-COVID window the IV-corrected expectations coefficient is positive, significant, and remarkably stable across the three measures: $\widehat{\psi}^{\mathrm{IV}} \in [0.29, 0.38]$, between 1.6 and 3.3 times the OLS estimate. The implied direction of OLS bias is downward, indicating that conventional fixed-effects regressions understate the strength of the regional expectations channel.

Second, the same pre-COVID IV specification drives the slope of the Phillips curve to zero. The unemployment coefficient under IV is statistically indistinguishable from zero in every measure. The apparent OLS Phillips curve is therefore largely driven by joint endogeneity.

Third, the post-COVID window (2021m3–2022m6) tells a strikingly different story. The IV expectations coefficient collapses to statistically insignificant levels, while the IV unemployment coefficient becomes large and strongly negative: -1.358 to -1.666 across the three measures, an order of magnitude steeper than any pre-COVID OLS estimate. The Phillips curve appears to have re-emerged in the post-pandemic inflation surge, even as the expectations channel attenuates – a pattern relevant to the post-COVID inflation debate (Coibion and Gorodnichenko, 2025; Benigno and Eggertsson, 2023; Beaudry et al., 2024).

Two near-term extensions follow naturally. Enriching the first-stage specification to include income, race, occupation, and employment status, with Bayesian shrinkage to stabilize estimates in sparse regions, would improve the precision and representativeness of the synthetic measures. Extending the analysis to firm-level expectations from the Survey of Firms' Inflation Expectations (SoFIE) and the Atlanta Fed's Business Inflation Expectations survey (Meyer and Sheng, 2024) would let us test whether the household-expectations channel documented here is complemented or dominated by the firm-expectations channel that Coibion and Gorodnichenko (2025) identify as central to price-setting.

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